

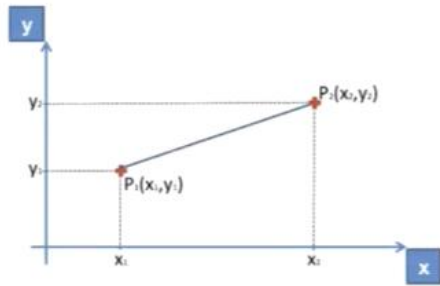
Feature Scaling

07 February 2026 18:13

Feature Scaling is a technique to standardize the independent features present in the data in a fixed range

Why do we need feature scaling?

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$$\text{Euclidean Distance between } P_1 \text{ and } P_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Let x be the no. of Salary and y be the no. of Age

Example: x_1 & y_1 are in row 2, x_2 & y_2 are in row 9

$$(x_2 - x_1)^2 = (83000 - 48000)^2$$

$$= 1225000000$$

$$(y_2 - y_1)^2 = (50 - 27)^2$$

$$= 529$$

We have age and CGP column, and when we use KNN (which is actually an algorithm where we simply calculate distance between two points),

Age	Salary	Purchase
50	83k	
27	48k	

Standardization

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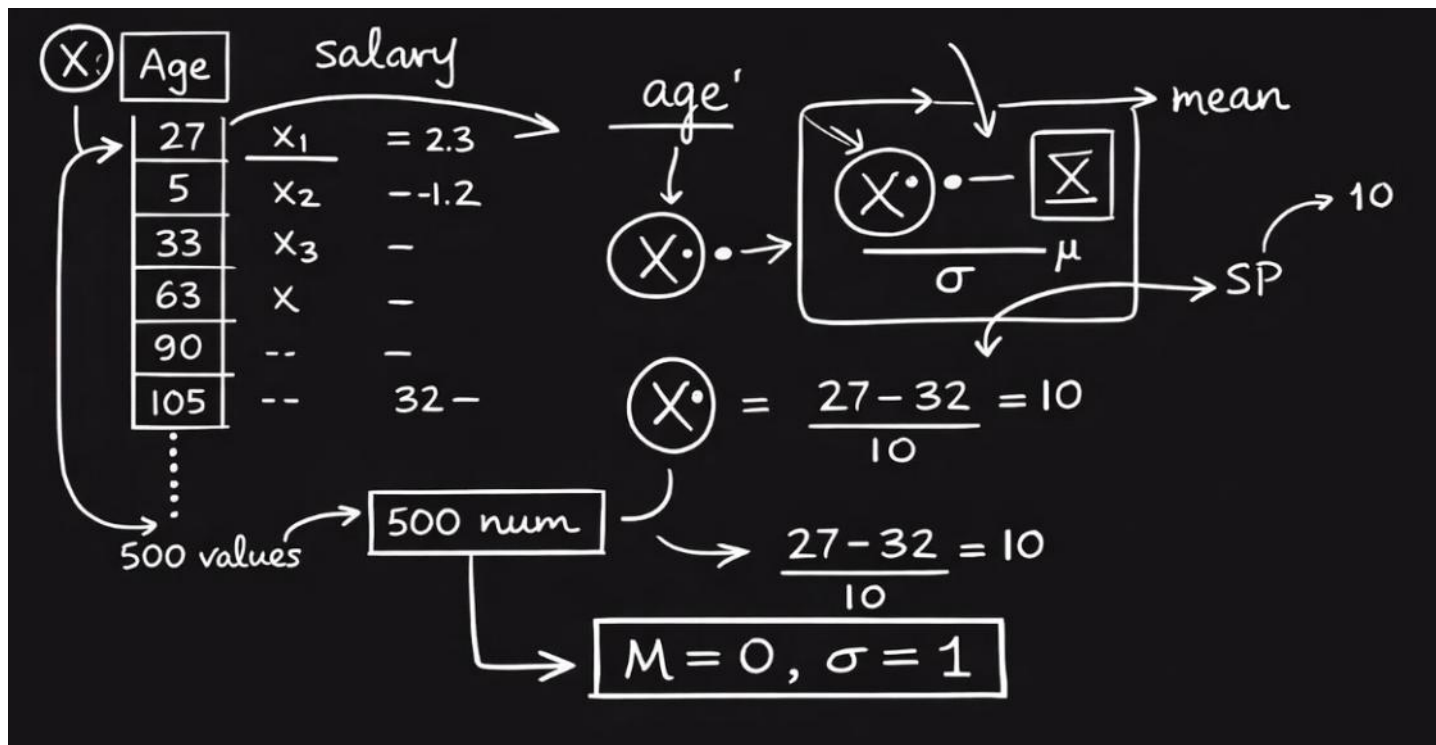
Z- Score Normalization

Original feature value Mean of the feature

$$z = \frac{x - \mu}{\sigma}$$

standard deviation of the feature

The rule for standardization is when you transform the column to a new column feature, then the feature, the mean for the new feature is always 0, and a standard deviation is 1.

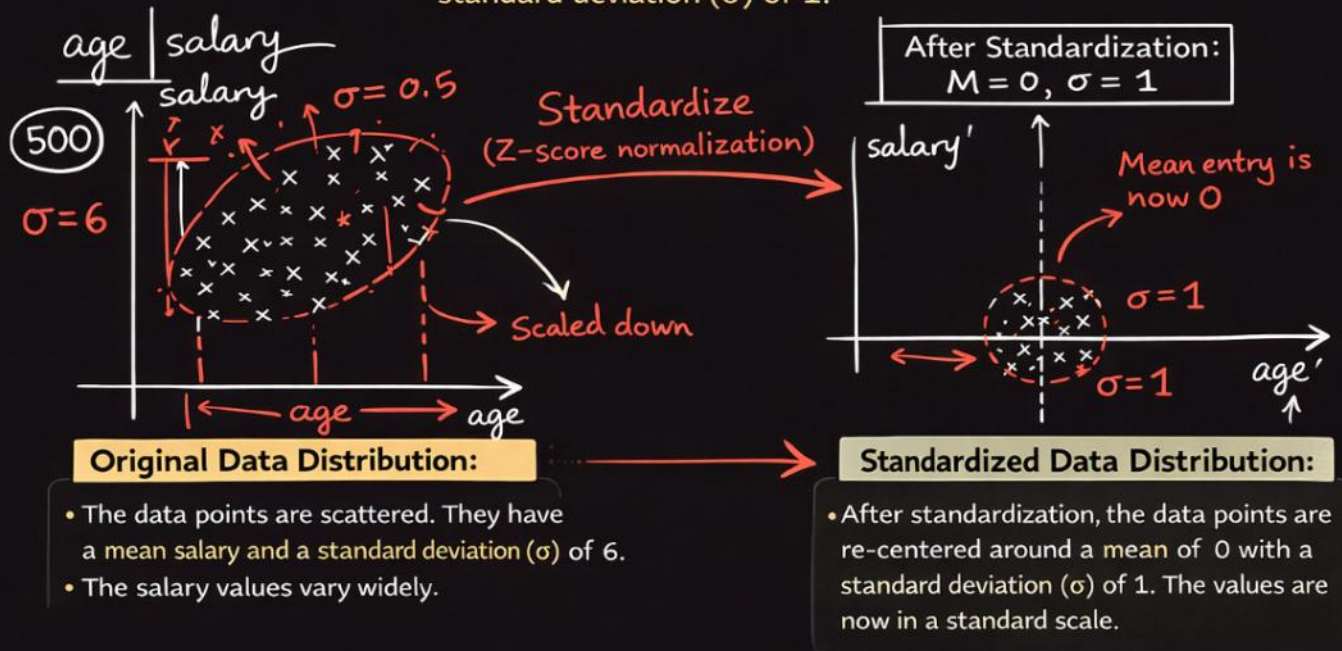


Why should we use standardization?

08 February 2026 09:46

Understanding Standardization

Standardization transforms the data to have a mean (M) of 0 and a standard deviation (σ) of 1.



When to use standardization?

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Algorithm(s)	Reason of applying feature scaling
1. K-Means	Use the Euclidean distance measure.
2. K-Nearest-Neighbours	Measure the distances between pairs of samples and these distances are influenced by the measurement units
3. Principal Component Analysis (PCA)	Try to get the feature with maximum variance
4. Artificial Neural Network	Apply Gradient Descent
5. Gradient Descent	Theta calculation becomes faster after feature scaling and the learning rate in the update equation of Stochastic Gradient Descent is the same for every parameter